

Question				Solution	Marks
3	(a)			The electromotive force of a source is the <u>electrical energy, converted from other forms of energy per unit charge.</u> transferred by the source in driving a charge round a complete circuit.	B1 B1
	(b)			The filament's resistance to increase with increasing p.d. When the temperature of resistor X increases, the <u>lattice ions gain more kinetic energy and vibrates more vigorously.</u> This results in the <u>more frequent collision between the conducting electrons and lattice ions</u> which reduces the mobility of the electrons (charge carriers).	A1 B1 B1
	(c)	(i)		Current passing through X and R must be the same. $V_X + V_R = 2.5 \text{ V}$ From graph: $I = 0.20 \text{ A}$	C1 C1 A1
		(ii)		$X = 3.3 \Omega$ $Y = 5.0 \Omega$	A1 A1

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4	(a)	(i)	$\text{e.m.f.} = (\Delta) B \times \Delta N / t$	C1